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Brunhes Epoch Geomagnetic Secular Variation on Marion Island: Contribution to Evidence for a Long-Term Regional Geomagnetic Secular Variation Maximum

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A paleomagnetic study of 30 lavas from Marion Island, situated on the Indian-Atlantic ridge yields a value for Brunhes epoch secular variation with confidence limits so broad that no distinction can be made with present geomagnetic field behavior. When combined with similar results from the nearby Crozet Islands, however, data for over 100 separate lavas become available for analysis, facilitating an estimate of regional secular variation with narrow confidence limits. The result, which suggests that Brunhes epoch geomagnetic secular variation in this region was (at the 95 percent confidence level) higher than at present, could be explained if the present secular variation maximum south of South Africa persisted during at least the past half million years. It is concluded that values of Brunhes epoch geomagnetic secular variation may depend not only on latitude, experimental methods, and coverage of the geological interval involved, but also on proximity to long term isoporic foci. Incidental results for the survey include detection of excursions attributable to non-dipole behavior in some instances, and possible dipole tilting in other cases.

1. Introduction

Latitude dependence of the angular standard deviation (S_t) of virtual geomagnetic poles (VGP) derived from paleomagnetic studies of lavas, when corrected for within-lava variations (to produce S_f) have been used to test geomagnetic field models for the past 0.69 million years in the Pacific (DOELL and COX, 1971, 1972), and Southern Indian Ocean regions (WATKINS and NOUGIER, 1973). The simple dipole wobble model of CREER *et al.* (1959) is excluded by the results, but distinction between the alternative models of IRVING and WARD (1964), which employs varying orientation but fixed magnitude non-dipole components superimposed on a fixed axial dipole, and Model C (DOELL and COX, 1972), which is based on present non-dipole field behavior plus dipole wobble, is not possible because of the difficulty in obtaining values of S_f with sufficiently narrow confidence limits.

Determination of S_f for selected parts of the earth's surface can in theory

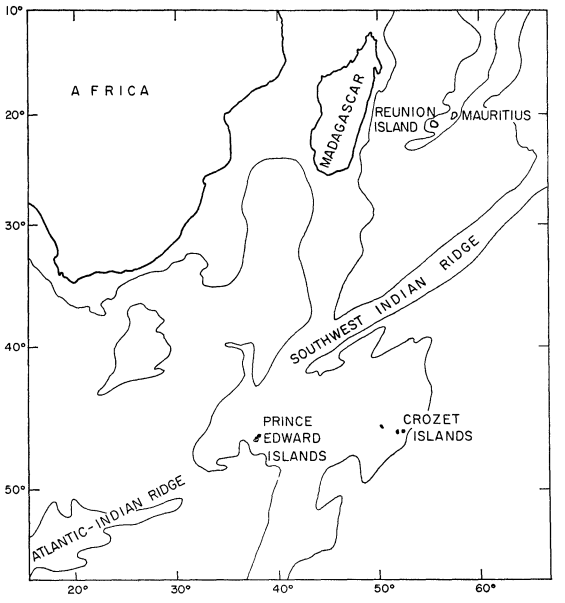


Fig. 1. Sketch map showing location of Prince Edward Islands (Marion and Prince Edward) and the Crozet Islands, with respect to mid-oceanic ridge system.

also be used to study the duration of non-dipole behavior. For example, the present geomagnetic secular variation minimum in the central Pacific (COX, 1962) has apparently persisted throughout the Brunhes epoch (DOELL and COX, 1971, 1972), although AZIZ-UR-RAHMAN and MCDOUGALL (1973) have suggested, on the basis of paleomagnetic data from Norfolk Island, that rapid accumulation of the lavas studied could also explain the low values of S_f. Similarly, MCELHINNY and MERRIL (1974) have questioned the long-term reality of the Pacific secular variation minimum. In contrast, no attempts have yet been made to investigate the possibility of persistence of secular variation maxima.

It is the purpose of this paper to present the results of a paleomagnetic survey of Marion Island (Figs. 1 and 2). As well as contributing data to the testing of geomagnetic field models by the latitude dependence of S_f, we shall test the possibility that present secular variation maxima have persisted for long periods, by combining the data from Marion Island with that already published by WATKINS *et al.* (1972) for Crozet (Fig. 1), since these relatively close islands are within the major isoporic focus south of South Africa (Fig. 3), which is at present changing at a rate of 130 nannotesslas per year. Combining the two results will not only provide a substantial time coverage for the Brunhes epoch for a restricted area, but also facilitate obtaining a result with narrow confidence limits.

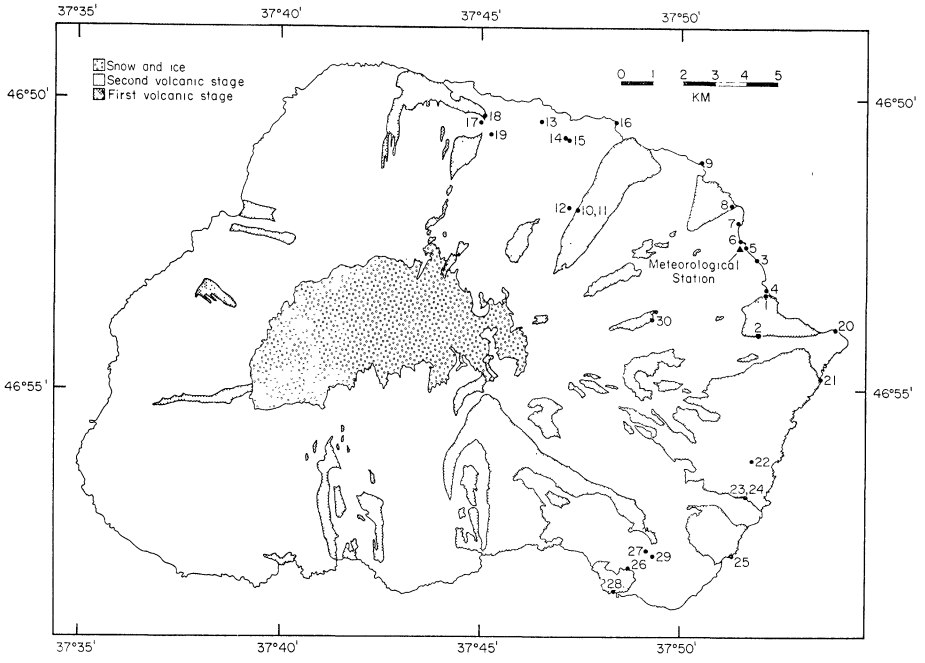


Fig. 2. Geological map of Marion Island, after VERWOERD (1971), showing sample site (lava) locations.

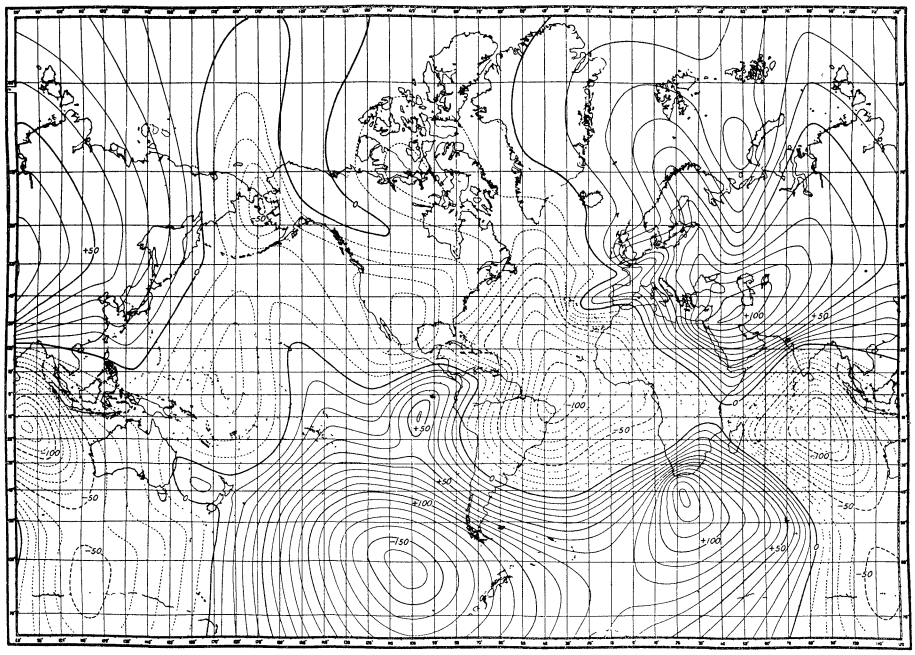


Fig. 3. Geomagnetic secular changes in nannotesslas per year, for the period 1940 to 1945 (from VESTINE *et al.*, 1947). A major isoporic focus or center of secular variation encloses the study area (Fig. 1).

2. Geology of Marion Island

Marion and Prince Edward Islands are exposed parts of a single volcanic platform on the northern flank of midoceanic ridge segment between the Atlantic and Indian oceans (Fig. 1), reaching elevations of 1203 and 672 m above sea level, respectively. Their combined area is 156 km^2, with Marion accounting for 70 percent of this total. The South African Biological and Geological Expedition to the islands in 1965 to 1966 produced a comprehensive series of studies of the islands (VAN ZINDEREN BAKKER *et al.*, 1971). Marion island is described by VERWOERD (1971) as being composed of an alternating series of explosive and effusive eruptions, divided into two stages separated by a glacial sequence. His geological map is reproduced in Fig. 2. The lavas are all alkali basalts or trachybasalts. The present surface is marked by radial fracturing, many small scoriatious cones and vents, and a small permanent snow cap in the center of the island. The results of chemical analyses by KABLE *et al.* (1971) show that Marion Island has strong affinities with other oceanic alkali basalts. Potassium-argon age determinations by MCDOUGALL (1971) suggests that the Older Series (Fig. 2) are between 300000 and 100000 years old, and that the Younger Series reach only about 23000 years in age. SNAPE and RETIEF (1971) have presented paleomagnetic data for seven samples from Marion Island, showing that only normal polarities were found.

3. Methods

Field. A total of 220 cores of 2.5 cm diameter were taken from 30 separate lavas, using a gasoline-powered portable drill. Sample sites are added to the geological map (Fig. 2). Ten sites were in the Older Series and twenty in the Younger Series. All cores were oriented, while still attached to outcrop, to a precision of $2\frac{1}{2}°$ using either a sun compass or geographic bearings taken from the available topographic map. Spacing between cores in each lava averaged 8 m. A site location and petrological description for each lava is available on request from the authors.

Laboratory. Specimens of 2.2 cm length were cut from all cores. Spinner magnetometers were used to measure the direction and intensity of remanent magnetism before and after the application of alternating magnetic fields of 100 and 200 oersteds. Final mean directions of remanent magnetism for each lava were computed using a minimum dispersion criterion (WATKINS *et al.*, 1972).

4. Results

Data for each lava are given in Table 1. The results are divided into those lavas with VGP's having latitudes less than, and more than 45°, respectively.

Table 1. Paleomagnetic data for Marion Island.

(a) Lavas with Low Latitude Virtual Geomagnetic Poles

No.	N	D	I	R	J	K	α_{95}	ϕ'	θ'
1	5(6)	279.8	−48.1	4.7210	0.381	14.3	20.9	293.4	27.3
6	8	82.1	−70.8	7.9486	14.8	136.1	4.8	169.5	40.7
7	6(7)	151.0	−54.2	5.7801	14.2	22.7	14.4	194.3	−4.4
12	7	73.5	−69.4	6.9383	22.5	97.2	6.2	164.0	44.4
16	7	83.5	−60.8	6.9478	18.2	114.9	5.7	155.9	33.0
21	8	89.5	−30.8	7.0353	2.30	7.3	22.1	139.1	12.4
23	7	67.5	−10.7	6.6746	5.43	18.4	14.4	114.8	19.2

(b) Analysis of Directions

No.	N	D	I	R	J	K	α_{95}	ϕ'	θ'
2	8	33.4	−82.4	7.8664	22.1	52.4	7.7	202.2	58.4
3	8	22.3	−70.0	7.3774	18.9	11.2	17.3	162.6	74.2
4	7	24.7	−66.9	6.9708	1.35	205.4	4.2	146.3	73.4
5	8	2.9	−74.2	7.9713	11.6	244.0	3.6	211.8	76.3
8	8	331.6	−69.8	7.8464	11.5	45.6	8.3	276.9	70.8
9	4(5)	8.3	−57.4	3.9364	30.3	47.2	13.5	75.4	79.2
10	6	23.2	−64.6	5.9212	5.93	63.5	8.5	135.0	74.1
11	4(5)	26.2	−45.0	3.8733	21.8	23.7	19.3	92.5	61.0
13	6(7)	353.3	−69.8	5.9243	24.7	66.1	8.3	248.0	82.1
14	7	2.0	−51.9	6.9123	18.6	68.4	7.3	44.8	75.5
17	8	349.9	−68.9	7.9869	18.9	532.4	2.4	264.4	81.5
18	6(8)	20.4	−61.5	5.5512	4.05	11.1	21.0	118.7	75.0
19	7(8)	359.1	−68.6	6.4367	22.0	10.7	19.4	224.1	85.0
20	8	2.8	−77.6	7.9577	11.7	165.3	4.3	214.5	70.6
22	6(7)	29.1	−80.8	5.9808	20.5	260.2	4.2	199.6	61.5
24	8	348.0	−50.5	7.8707	19.3	43.9	8.4	3.3	71.8
25	6	65.0	−72.2	5.9752	10.1	201.7	4.7	167.6	50.4
26	8	38.0	−54.6	7.9712	10.4	242.7	3.6	119.1	59.4
28	8	15.9	−65.9	7.9627	15.7	187.8	4.1	140.2	79.2
29	7	11.5	−59.3	6.8452	16.8	38.8	9.8	92.2	79.3
30	5(7)	356.8	−71.8	4.8466	5.86	26.1	15.3	227.9	80.0

(c) Analysis of Virtual Geomagnetic Poles

No.	R	K	α_{95}	ϕ'	θ'
2	7.5323	15.0	14.8	202.3	58.6
3	6.6673	5.3	26.7	174.6	72.9
4	6.9395	99.1	6.1	147.4	73.5
5	7.9193	86.8	6.0	211.4	76.1
8	7.6675	21.1	12.4	274.6	71.7
9	3.8692	22.9	19.6	76.5	79.1
10	5.8648	37.0	11.2	139.3	74.3
11	3.8668	22.5	19.8	95.0	61.3
13	5.7829	23.0	14.3	247.8	81.9
14	6.9199	74.9	7.0	46.6	76.0
17	7.9710	241.4	3.6	263.5	81.5

(continued) Table 1.

No.	R	K	α_{95}	ϕ'	θ'
18	5.3505	7.7	25.8	135.0	73.7
19	5.8076	5.0	29.9	215.0	79.3
20	7.8603	50.1	7.9	214.8	70.4
22	5.9350	76.9	7.5	200.1	61.3
24	7.7960	34.3	9.6	4.6	72.4
25	5.9302	71.6	8.0	167.9	50.6
26	7.9488	136.6	4.8	119.4	59.3
28	7.9178	85.2	6.0	142.0	79.2
29	6.7976	29.6	11.3	91.4	80.6
30	4.6230	10.9	24.3	222.9	78.9

No.=Flow number (see Fig. 2). N=number of separate cores. Number in bracket is original number of cores prior to rejection of totally aberrant results. D and I=declination and inclination of remanent magnetism after treatment in alternating magnetic fields; D is in degrees east of north; I is in degrees with respect to the horizontal (−=upwards). R=resultant vector, applying unit vector per specimen. J=intensity of magnetization in emu gm^{-1} to 10^4. $K=N-1/N-R$. α_{95}=semi-vertical angle of 95 percent confidence cone (FISHER, 1953). ϕ' and θ'=longitude and latitude of virtual geomagnetic pole: ϕ' is in degrees east; θ' is in degrees north.

Totally abberant directions of remanent magnetism were found in a total of 11 specimens (Table 1). These data, suspected to result in part from core orientation and marking problems in the field, are subjectively rejected in later compilation.

5. Discussion

Magnetic Properties. Stability and intensity of magnetization are highly variable within almost all lavas, as might be expected in view of commonly observed within-lava variation (WATKINS and HAGGERTY, 1967). Two of the lavas, (Nos. 15 and 27) are highly unstable throughout, and no mean stable direction of remanent magnetism could be determined. For those lavas with lower latitude VGP (Table 1) the Older Series units have systematically lower J values, consistent with spontaneous decay of J with time; but the highly variable J values for the larger group of lavas (with higher VGP latitudes) shows that there are no overall systematic variations of J with age.

Lower Latitude Virtual Geomagnetic Poles. Seven of the lavas sampled possess VGP latitudes less than 45° (Table 1). These are considered separately, since they are unlikely to represent normal geomagnetic behavior. These VGP are, with one exception, in reasonable proximity to each other. The simplest interpretation of these data would, therefore, be that a single departure of the virtual geomagnetic pole to low latitudes has been recorded by a series of eruptions closely spaced in time, but since lava Nos. 21 and 23 are part of the Older Series (and are, therefore, older than 100000 years) and lava Nos. 6, 7, and 12 are part of the Younger Series, such an explanation is not possible. As in the

case of a similar study of Brunhes epoch lavas on Amsterdam Island in the southeastern Indian Ocean (WATKINS and NOUGIER, 1973), the results require that a geomagnetic excursion has occurred with similar directional character-

Table 2a. FISHER (1953) statistical analyses of directions and VGP.

Island	B	N	D	I	R	K	α_{95}	θ'	ϕ'
					Analysis of directions				
Marion	21	191	11.9	−67.1	20.5147	41.2	5.0	81.6	152.4
East (Crozet)	43	185	7.5	−61.7	41.6546	31.2	4.0	83.6	111.1
Possession (Crozet)	37	181	1.5	−63.1	36.0963	39.8	3.8	87.9	81.5
All data combined	101	557	5.4	−62.2	97.5816	29.3	2.6	85.1	97.7
					Analysis of VGP				
Marion	21	191			20.0355	20.7	7.2	80.4	166.7
East (Crozet)	43	185			40.4947	16.8	5.5	84.7	125.4
Possession (Crozet)	37	181			35.4498	23.2	5.0	88.7	122.3
All data combined	101	557			95.3254	17.6	3.4	86.2	123.7

B=No. of separate sites. N=total number of specimens. D and I=mean declination and inclination of remanent magnetism (D in degrees east of north; I in degrees with respect to horizontal, negative being upward). R=resultant vector. $K=N-1/N-R$. α_{95}=semi-vertical angle of 95% confidence core. θ' and ϕ'=latitude and longitude of mean virtual demagnetization and minimum dispersion computation for each site; rejection of data for selected specimens and for sites with VGP latitudes less than 45°. All data are for unit vector per site.

Table 2b. Angular standard deviation of VGP.

Island	S_t	Kt	Kw	Ka	Kf	S_f	Su	Si
		Computation of S_f with respect to Mean VGP*						
Marion	18.2	19.9	20.2	2100	23.5	16.7	21.1	13.8
East (Crozet)	20.1	16.3	43.9	2100	18.0	19.1	22.3	16.8
Possession (Crozet)	17.1	22.5	54.9	2100	24.9	16.2	19.1	14.3
All combined	19.7	16.9	29.9	2100	19.3	18.5	20.5	16.9
		Computation of S_f with respect to Present Spin Axis						
Marion	20.3	15.9	20.2	2100	18.2	19.0	24.0	15.7
East (Crozet)	20.7	15.3	43.9	2100	16.8	19.8	23.1	17.4
Possession (Crozet)	17.2	22.3	54.9	2100	24.7	16.3	19.2	14.3
All combined	20.1	16.3	29.9	2100	18.5	18.9	20.9	17.2

S_t=total dispersion where $S_t^2=(1/B-1)\sum_{i=1}^{\beta}(\delta_i)^2$ where δ_i=angular difference between the i-th site VGP and the reference pole, and B=number of sites. $Kt=2(S_t)^{-2}$. Kw=within-lava precision parameter for VGP, $Kw=\sum_{i=1}^{N}(N_i-1)/(N_i-R_i)$ where N=number of specimens in i-th lava, and R=resultant vector for i-th lava, applying unit vector per specimen VGP. Ka=precision parameter for effect of magnetic anomaly existing during lava emplacement at the site (see DOELL, 1972 and ELLWOOD *et al.*, 1973 for discussion). Kf=precision parameter reflecting ancient geomagnetic field variation: $(1/Kf)=(1/Kt)-(1/\bar{N}KW)-(1/Ka)$ where $\bar{N}$=average number of specimens per site. $S_f=(2/Kf)^{1/2}$. Su and Si=upper and lower confidence parameters for S_f (after COX, 1969). All data for bodies with VGP latitudes ≧45°.

* See Table 2a.

istics on at least two separate occasions during the Brunhes epoch. Some of the other excursions reported for the Brunhes epoch (WATKINS, 1973; WATKINS and NOUGIER, 1973; ELLWOOD *et al.*, 1973) provide VGP in the vicinity of those found on Marion Island. For example, five lavas in Amsterdam Island record VGP's in the vicinity of that for lava No. 1 (Table 1a); one excursion found on Reunion Island has a VGP close to those for lava Nos. 6, 12, and 16 (Table 1a). This suggests but does not prove that dipole tilting could be the cause of these particular excursions. Thus, for three of the Marion Island lavas the low latitude VGP are interpreted most likely to represent probable repetition of non-dipole activity during a period of diminished dipole intensity, and in four cases, possibly dipole tilting. The results are also relevant to the model of CREER and ISPIR (1970), who invoke a quasi-permanent inclined minor dipole directed toward the central Pacific to explain the similarity of several VGP paths representing geomagnetic polarity transitions between opposite polarities in lava sequences as old as Miocene. The VGP positions for the low latitude poles are in five of the seven cases, in the Pacific area. This may be simple coincidence, or could indicate a detection of the limited period when the minor inclined dipole of CREER and ISPIR (1970) became dominant. This form of analysis of the mechanism involved in the production of low latitude VGP's on Marion Island is weak, since the proximity of Brunhes VGP's from other areas could as well be the result of a random process.

Secular Variation. The value of S_f derived from the Marion Island lavas with VGP latitudes higher than 45° (Table 2) is within the 95 percent confidence limits (COX, 1969) similar to persent geomagnetic field values for the same latitudes (Fig. 4a). The wide confidence limits prevent further conclusions.

Data from Possession Island and East Island in the Crozet group, which are only 15° of longitude from Marion Island (Fig. 1) have been added to the Marion Island results, to produce a value of S_f for the region. This becomes only the third region for which data from over 100 separate bodies are available to determine S_f for the Brunhes epoch, the others being the Hawaiian and Galapagos islands (DOELL and COX, 1972). It can be seen (Fig. 4b) that the regional value of S_f is higher than average present geomagnetic field behavior for the latitudes involved. This would require mean variations of D and I which increase by only five to ten percent above those required to produce S_f values for the present field. In Fig. 5 we show the variation in D and I which *could* produce the present S_f value of 15.7 needed (used in Fig. 4b). Added to this is a D and I variation which *could* produce the observed Brunhes epoch value of 19.0 for Marion Island (Fig. 4a). We stress that these curves are idealized, because although there appeared to be regular variation of D and I for the London observatory (which is at northern latitude almost identical to the southern one of Marion) since the late sixteenth century (Fig. 5), the long-term D and I variation is by no means regular, as shown by AITKEN (1970) using archeomagnetic

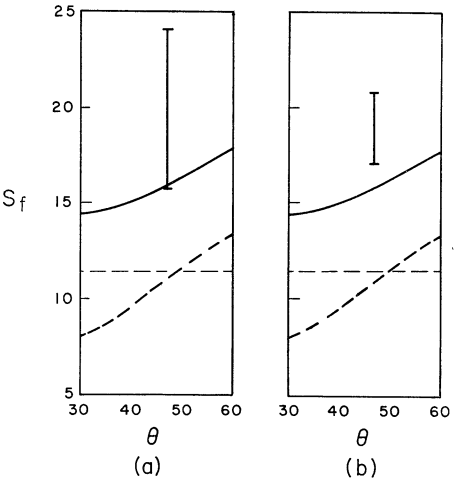


Fig. 4. Corrected angular dispersion of virtual geomagnetic poles (S_f) plotted as 95 percent confidence limits only. Horizontal line=effect of simple dipole wobble model for geomagnetic field. Value of 11° for Brunhes epoch from DOELL (1969). Dashed line=effect of present non-dipole field, averaged between hemispheres. Upper curved line=combined effect of present non-dipole field and dipole wobble (DOELL and COX, 1972). θ=latitude. (a) S_f for Marion Island lavas (Table 2b). (b) Regional value of S_f from data for Marion and for Crozet Islands (Table 2b).

data. Inspection of Fig. 5 incidentally shows that to provide a value of S_f of 15.7, the range of D and I fluctuations which might be required to arrive at this average value would be very large: the London data for the past 1000 years would, if sampled at 100 year intervals, provide an average S_f value much smaller than 15.7. This suggests that it may be legitimate to include data with VGP latitudes less than 45° (DAGLEY and WILSON, 1971) while analysing the Brunhes dipole field.

The preferred interpretation for the result shown in Fig. 4b is that during the Brunhes epoch, the southeastern Atlantic to southwestern Indian Ocean region was an area of anomalously high geomagnetic secular variation. Alternative interpretations of the result can only be in the categories of experimental deficiences or insufficient sampling of the secular variation spectrum. The former possibility is unacceptable, since over 100 units are involved, with an average of 5.6 separate cores per lava, and unstable components have been minimized in all cores. The latter possibility is similarly unlikely, since potassium-argon dating shows that the last 0.3 million years is represented on Marion Island, and the lower Brunhes is represented on Possession Island. The Crozet group is a totally separate volcanic edifice (GUNN *et al.*, 1970) to that of Marion and Prince Edward. Thus substantial time coverage of the Brunhes epoch has

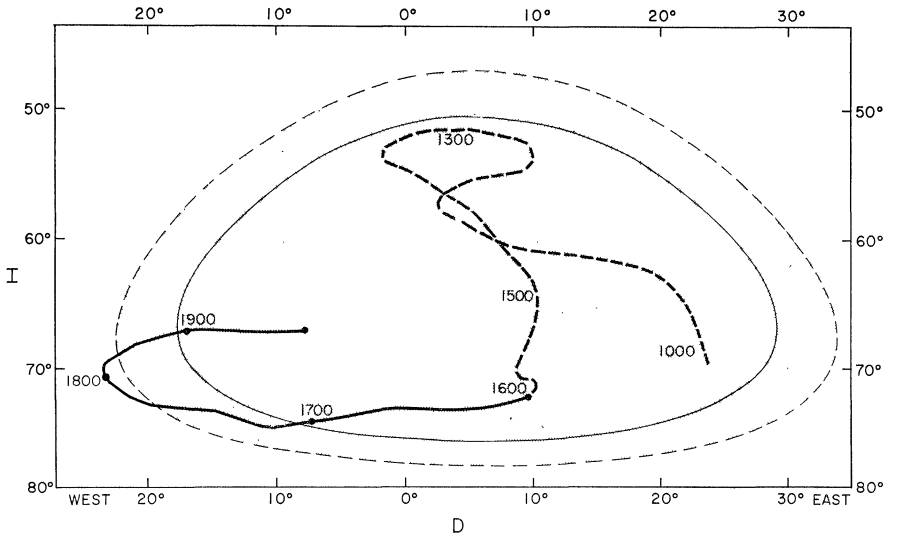


Fig. 5. This shows the magnitude of the increase in average Declination (D) and Inclination (I) necessary to elevate S_f to the observed value of 19.0 for Marion Island (dashed closed figure) compared to the value for the present geomagnetic field (solid closed Figure). The variation of D and I values for England since 1000 A.D., using observatory and archeomagnetic data (AITKEN, 1970) are added. Note: I is negative for normal polarity at Marion Island and positive in England, which is at the same north latitude as Marion.

been obtained. In addition, the rejection of data for lavas with VGP latitudes less than 45° probably contributes to computed S_f results lower than real values. It is proposed that maximum prospects exist for fulfillment of both experimental and age coverage requirements for this particular experiment.

We conclude that a Brunhes epoch regional geomagnetic secular variation anomaly has been demonstrated for the region. A necessary if insufficient condition has therefore been met for confirmation of the existence of a standing non-dipole field component. This would mean that the present isoporic focus is in an area with a standing non-dipole component. The condition is insufficient because other explanations are possible to provide results consistent with division of the non-dipole field into standing and drifting components. For example, the non-dipole behavior which results from analysis of the present field as shown in Fig. 4 is computed by averaging the non-dipole field over 10° longitudinal intervals at each latitude in both hemispheres, and averaging between hemispheres. The result in Fig. 4b could therefore be explained if non-dipole field activity was not hemispherically symmetrical on average during the Brunhes; but, rather, was greater in southern latitudes.

YUKUTAKE and TACHINAKA (1968) have shown that the present non-dipole field is divisible into drifting and standing components, unlike the earlier concept of a totally drifting non-dipole field (BULLARD *et al.*, 1950). If this model

is valid, then areas of present secular variation maxima (or isoporic foci) could perhaps have been similarly active in the past, particularly if the standing component also undergoes time variations. This would require generation of the non-dipole component in the lower mantle, or a core-mantle effect based on mantle properties since the core moves with respect to the mantle. It would also require that the standing components are related to the present isoporic focus over Marion and Crozet (Figs. 1 and 3). This is contrary to analysis of YUKUTAKE and TACHINAKA (1969), who show that the present isoporic focus involved is attributable almost completely to a drifting component. It remains to be seen if further analyses of the present field provide confirmation that this major isoporic focus is totally without any standing component (YUKUTAKE, 1973).

One test of the proposition that present isoporic foci reflect in part a standing component of the non-dipole field during the Brunhes epoch would be a correlation between anomalously high S_f values and the position of the foci. The number of computations of S_f for the Brunhes epoch is not large. DOELL and COX (1972) have published results for the Galapagos Islands, Hawaii, the Western U.S.A., New Zealand, France, Alaska, Iceland, and Antarctica. To these results can be added those from the Azores and the Canary Islands (ELLWOOD *et al.*, 1973): Reunion (WATKINS, 1973); The Comores and Amsterdam Island (WATKINS *et al.*, 1972; WATKINS and NOUGIER, 1973); and Easter Island (ISAACSON *et al.*, 1973). Of all these results only those for Alaska and Reunion are similar to the result for Marion Island and Crozet (Fig. 4b), in that they are significantly higher at the 95 percent level than the present average value for the site latitude involved. While a weak isoporic focus (of -50 nannotesslas per year) does exist directly over the Alaska site locations and Reunion is on the western flank of the major northern Indian Ocean isoporic focus (Fig. 3), the possible meaning of the correlations is weakened because the Galapagos sites (where the value of S_f derived is not anomalously high) are also very close to another minor isoporic focus. On the other hand, none of the other ten results (which all provide S_f values comparable to present values) are close to isoporic foci. Thus, while far from sufficient to prove the point, on balance there exists (on the one hand) a relationship between anomalously high S_f values and positions of isoporic foci, and (on the other hand) S_f values which are consistent with present geomagnetic field behavior when the sites involved are in regions removed from isoporic foci. This does not mean that the geomagnetic field would not be averaged out to the spin axis, since as Table 2(a) shows, the present spin axis is still represented by the mean directions of remanent magnetism. A time-varying standing component would also be required. The conclusion is that secular variation during the Brunhes epoch will not only depend on latitude, but also possibly on proximity to a long-term isoporic focus. It follows that a standing component of the non-dipole field would be required to originate in the lower mantle, or to be related to interaction between the core and a long

term feature of the lower mantle, since relative motion between the core and mantle would otherwise ensure longitudinal drift of the entire non-dipole field.

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